

**PREDICTING CYBERSECURITY THREAT SEVERITY
USING MACHINE LEARNING:
A CRITICAL EVALUATION OF SYNTHETIC DATASET
VALIDITY IN BUSINESS ANALYTICS**

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Declaration

This is to certify that:

- i. The portfolio comprises only my original work;
- ii. AI technologies (e.g. ChatGPT) have not been used in the writing of the portfolio dissertation;
- iii. No portion of the work referred to in the dissertation has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

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Abstract

Machine learning is now established as the dominant analytical approach to cybersecurity threat detection, with peer-reviewed studies reporting classification accuracies exceeding 95 per cent on documented benchmark datasets. This study set out to apply that approach to the prediction of cybersecurity threat severity in organisations, using a publicly available synthetic dataset of 50,000 threat scenarios obtained from an open data repository.

Following the CRISP-DM framework, four regression algorithms were applied to four candidate target variables under five-fold cross-validation. All sixteen configurations returned negative coefficients of determination, indicating that no model outperformed the naive strategy of predicting the sample mean. Systematic diagnostic testing then excluded model capacity, target specification, data volume, feature availability and problem formulation as explanations, and a permutation test established that models fitted to genuine target values performed no better than those fitted to randomly shuffled labels.

Diagnostic analysis identified three characteristics accounting for this outcome. Twelve of thirteen numeric variables proved statistically indistinguishable from uniform random distributions under two independent goodness-of-fit procedures. Across 120 pairwise correlations none exceeded 0.0112, and no categorical predictor explained more than 0.012 per cent of variance. Most significantly, a composite control score presented as a summary of its own components was explained by them to a degree of 0.006 per cent — difficult to reconcile with genuinely derived data.

The study concludes that the dataset shows evidence of internal inconsistency and is unfit for predictive modelling, and that its provenance could not be verified retrospectively. These findings extend an established tradition of cybersecurity dataset critique, differing from prior work in degree rather than kind. The principal contribution is a five-stage synthetic dataset validation protocol, executable in under one hour, comprising

provenance audit, distribution screening, correlation screening, internal consistency testing and permutation-based signal verification. Recommendations are advanced for research practice, for repository governance and for the treatment of dataset validation as a discrete phase of the analytical workflow.

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1. Introduction

1.1 Background and Context

Cybersecurity threats represent one of the most significant operational risks facing modern organisations. The financial cost of cyberattacks has risen sharply in recent years, with the average cost of a data breach reaching record levels alongside substantial reputational and regulatory consequences (IBM Security, 2023). Despite sustained investment in security infrastructure, many organisations remain reactive in posture, responding to incidents after damage has occurred rather than anticipating and prioritising threats beforehand.

Machine learning has emerged as the principal analytical response. Zhang et al. (2022) document consistent success across industrial and network environments, and ensemble methods have proved particularly effective: Dasari and Kaluri (2024) report 99.77 per cent accuracy applying gradient boosting to network attack data, whilst Mehmood et al. (2023) achieve 97 per cent on insider threat detection. The analytical toolkit is mature and widely available.

1.2 Problem Statement

That consensus rests on a condition receiving comparatively little scrutiny: the quality of the data to which these methods are applied. Sambasivan et al. (2021) describe a culture privileging model development over data work, producing data cascades — compounding downstream failures originating in unaddressed upstream problems. Gebru et al. (2021) argue that absent standardised documentation, consumers cannot assess fitness for purpose. Within cybersecurity, Tavallaee et al. (2009) demonstrated that a benchmark underpinning two decades of research contained artefacts materially inflating reported performance.

Public repositories have simultaneously made analysis-ready datasets freely available. This accessibility is a genuine benefit, but absent any provenance requirement, datasets

of arbitrary quality may enter the research pipeline carrying an unearned appearance of legitimacy. The consequences for cybersecurity, where analytical outputs inform the allocation of defensive resources, warrant examination.

1.3 Research Question, Aim and Objectives

This study is guided by the following research question: to what extent can machine learning techniques accurately predict the severity of cybersecurity threats in organisations, and which risk factors most significantly influence that severity?

The aim was to develop and evaluate machine learning models predicting cybersecurity threat severity and to identify the key risk factors driving organisational vulnerability, in support of proactive, evidence-based security decision-making. Five objectives supported this aim: to critically review the relevant literature; to prepare and explore the chosen dataset; to develop and compare multiple predictive models; to identify the variables most strongly associated with threat severity; and to evaluate model performance and its practical implications.

In pursuing these objectives the investigation encountered a result that redirected its emphasis. No model, on any candidate target, achieved performance exceeding that of predicting the sample mean. Two responses were available at that point: to continue adjusting features and hyperparameters until some positive result emerged, or to treat the failure itself as the object of investigation. The second was chosen, on the reasoning that a model tuned into apparent success on data containing no signal would be worse than no model at all. The original objectives were retained and are answered in Chapter 6; a sixth was added, namely to establish why prediction was not possible and to derive from that a procedure others could apply.

1.4 Contribution

The study makes three contributions. Empirically, it provides five independent lines of evidence that a publicly available cybersecurity dataset does not support predictive modelling, and identifies the specific properties responsible. Methodologically, it proposes a five-stage validation protocol executable in under one hour, intended for use before analytical effort is committed to an undocumented dataset. Substantively, it demonstrates that in a domain where analytical outputs inform the allocation of defensive resources, conventional data quality checks can be satisfied in full by data that is nonetheless unusable.

1.5 Report Structure

Chapter 2 reviews the literature across four themes: machine learning performance on authenticated datasets, benchmark dataset construction, data quality and its consequences, and existing critiques of cybersecurity datasets. Chapter 3 sets out the research design and analytical procedure. Chapter 4 reports the empirical findings. Chapter 5 interprets them, situates them within the literature and advances the validation protocol. Chapter 6 concludes against the objectives and offers recommendations.

2. Background and Literature Review

This chapter reviews the literature informing the study across four themes. The first establishes the performance that machine learning achieves on authenticated cybersecurity datasets, providing the benchmark against which this study's results are measured. The second examines how properly constructed benchmark datasets are built and documented. The third addresses the wider literature on data quality and its downstream consequences. The fourth reviews existing critiques of cybersecurity datasets, establishing the tradition within which this study is positioned.

2.1 Machine Learning in Cybersecurity Threat Detection

The application of machine learning to cybersecurity has been driven by the limitations of signature-based detection, which relies on predefined rules and performs poorly against novel attacks. Zhang et al. (2022) provide the most comprehensive recent survey, examining twenty deep learning applications to cyber-physical system security and proposing a six-step methodology spanning scenario analysis, attack identification, problem formulation, model construction, data acquisition and evaluation. Notably, the authors attribute the strong performance observed partly to the availability of high-quality public datasets — an observation whose corollary this study examines. Their survey is confined to deep learning architectures and binary attack detection, addressing neither interpretable ensemble methods nor the prediction of threat severity.

Ensemble methods dominate applied work on structured cybersecurity data. Dasari and Kaluri (2024) evaluate a hierarchical framework of XGBoost, LightGBM, CatBoost, random forest and decision tree on the CICIDS2017 benchmark, applying LASSO feature selection and SMOTE oversampling with hyperparameter optimisation; LightGBM achieved 99.77 per cent accuracy. Their study demonstrates that gradient boosting variants consistently outperform simpler classifiers, and establishes tuning and imbalance correction as standard pipeline components.

Mehmood et al. (2023) apply random forest, AdaBoost, XGBoost and LightGBM to insider threat detection in cloud environments using a customised CERT extract, with LightGBM attaining 97 per cent accuracy and random forest a false positive rate of 0.78 per cent. The study's principal methodological value lies in its comparative design — four classifiers on identical data under identical conditions — an approach adopted here. The authors acknowledge that reliance on a single customised source constrains generalisability.

Yeboah-Ofori et al. (2021) integrate cyber threat intelligence with machine learning to predict supply chain attacks, achieving an F-score of 0.85. Their contribution is conceptual as much as technical, demonstrating that structured threat intelligence attributes can be operationalised as model inputs. Karunamurthy et al. (2025) report 95.59 per cent accuracy using federated learning on a benchmark IoT dataset, extending the approach to distributed settings whilst preserving privacy.

A parallel literature addresses interpretability. Gaspar, Silva and Silva (2024) apply LIME and SHAP to a black-box intrusion detection model, going further than most explainability studies by validating the explanations themselves through perturbation analysis. They demonstrate that specific features consistently mislead predictions — information of direct operational value to analysts.

Collectively this work establishes a clear expectation: ensemble machine learning applied to structured cybersecurity data reliably extracts substantial predictive signal, with reported performance clustering between 95 and 99 per cent.

2.2 Benchmark Dataset Construction and Documentation

The datasets underpinning these results are not incidental to them. Moustafa and Slay (2015), introducing UNSW-NB15, document the originating institution, traffic generation tool, capture volume, feature extraction utilities and attack taxonomy. Their

subsequent paper (Moustafa and Slay, 2016) subjects the dataset to statistical evaluation against KDD99 — an exercise in self-validation anticipating the diagnostic approach adopted here.

Sharafaldin et al. (2018), introducing CICIDS2017, are more prescriptive still, specifying eleven criteria that a valid intrusion detection dataset must satisfy, including realistic background traffic, complete capture, labelled ground truth, documented attack diversity and defined feature sets. Their framework establishes that standards for dataset construction in this domain are neither absent nor obscure.

The most influential general framework is that of Gebru et al. (2021), whose proposal for standardised datasheets requires every dataset to document its motivation, composition, collection process and intended uses. Their central argument is that documentation is the mechanism by which a consumer assesses fitness for purpose; absent it, that assessment cannot be made. The framework has been widely adopted in principle but unevenly in practice, particularly outside curated repositories.

2.3 Data Quality and Its Consequences

A broader literature examines what follows when data quality is neglected. Sambasivan et al. (2021), studying high-stakes AI deployments, introduce the concept of data cascades: compounding downstream failures originating in unaddressed upstream problems. Their central finding is cultural — practitioners undervalue data work relative to model development, and the resulting problems surface late and expensively.

Canbek (2022) addresses the issue methodologically, arguing that quantitative evaluation of feature distributions should precede model development, framing the matter in garbage-in-garbage-out terms. Geiger et al. (2020) provide empirical evidence of scale, finding that many machine learning papers fail to report where training data originated.

Paullada et al. (2021) describe how inadequately documented datasets create faulty foundations for subsequent research.

Within cybersecurity specifically, Tran et al. (2022) argue for formal data curation and quality evaluation as a prerequisite to intrusion detection modelling. Arp et al. (2022), writing in a leading security venue, identify ten recurrent pitfalls in machine learning security research — among them sampling bias, spurious correlation, inappropriate baselines and inadequate validation — and observe that such failures frequently escape detection precisely because reported performance metrics appear satisfactory.

2.4 Critiques of Cybersecurity Datasets

The critical evaluation of cybersecurity datasets constitutes an established research tradition. Its foundational instance is Tavallaee et al. (2009), whose statistical analysis of KDD Cup 99 identified extensive record redundancy and demonstrated that these artefacts inflated reported classifier performance. The paper has been cited several thousand times and prompted construction of the NSL-KDD variant.

Subsequent work extended this critique. Al-Tobi and Duncan (2018) conducted a forensic review of generation faults in the same dataset; Siddique et al. (2019) argued that reliance on a flawed benchmark distorted intrusion detection research over two decades, framing the issue as one of field integrity rather than individual study validity; and Sapre et al. (2019) demonstrated that classifier performance is partly an artefact of dataset construction.

This literature establishes both the legitimacy and the value of dataset critique as a research contribution. It is notable, however, that the flaws documented in this tradition are consistently subtle: redundancy, class imbalance and sampling artefacts that inflate performance whilst leaving genuine underlying structure intact. Models trained on KDD Cup 99 did learn something, albeit less than reported.

2.5 Synthesis and Research Gap

Three observations follow. Ensemble machine learning applied to authenticated cybersecurity datasets reliably achieves high performance, establishing a clear expectation against which any new application may be measured. Standards for dataset construction and documentation exist and are widely cited, yet compliance outside curated repositories is unenforced. And dataset critique is an established form of contribution, but existing instances address subtle flaws in datasets that nonetheless contain genuine structure.

Two gaps emerge. The literature reviewed addresses attack detection almost exclusively, with comparatively little attention to the prediction of threat severity once a threat is identified — the task with the most direct bearing on operational resource allocation. More significantly, no existing study examines a dataset that contains no exploitable structure whatsoever, nor offers a systematic procedure by which such a dataset might be identified before analytical effort is committed to it. This study addresses both.

3. Design and Methodology

3.1 Research Design

The study adopts a quantitative, exploratory design structured according to CRISP-DM, which organises analytics projects into business understanding, data understanding, data preparation, modelling, evaluation and deployment. It was selected as an industry-recognised standard and for the separation it maintains between data understanding and modelling — a distinction that proved consequential here.

The design evolved during execution. The original specification was a predictive modelling study. When initial modelling produced no performance, the design expanded to incorporate a systematic diagnostic phase in which alternative explanations were tested and excluded in sequence before any conclusion was drawn about the data. This phase is reported as integral to the methodology rather than a departure from it, since the validity of the conclusions rests upon its thoroughness.

3.2 Data Source

The data comprises a publicly available cybersecurity threat intelligence dataset obtained from an open data repository, containing 50,000 records across 22 columns and representing simulated cyberattack scenarios. The dataset is secondary data and contains no personally identifiable information.

Retrospective attempts to relocate the original repository listing were unsuccessful; the uploading party, publication date, licence terms and generation methodology could not therefore be verified. This limitation is reported as a finding in its own right and examined in Section 5.3. The dataset represents simulated rather than operational data, though simulated datasets are widely used in this field owing to the sensitivity of genuine threat intelligence (Zhang et al., 2022).

3.3 Data Preparation

Initial inspection established record and column counts, data types, cardinality and missingness for every field. This inspection revealed that seven columns contained nested JSON objects and arrays rather than scalar values. Four of these — `asset_count`, `business_context`, `existing_controls` and `prediction_results` — encapsulated multiple numeric fields within single text strings.

These structures were parsed using regular expression extraction in R, recovering eleven additional numeric variables and expanding the analytical feature space from 22 to 33 columns. Categorical variables were converted to factors. One variable, `historical_incident_count`, was found to be constant across all records and was excluded from analysis on the grounds of zero variance. No imputation was required, as no missing values were present.

3.4 Analytical Procedure

The analysis proceeded in five stages. Descriptive statistics and distributional analysis characterised each numeric variable, with uniformity assessed through two independent goodness-of-fit procedures: the one-sample Kolmogorov–Smirnov test against the relevant uniform distribution, and a chi-square test across twenty equal-width bins. Two tests were employed because the Kolmogorov–Smirnov procedure assumes continuous data without ties, an assumption violated by several variables.

Association analysis computed Pearson correlations across all numeric pairs, supplemented by one-way analysis of variance with eta-squared effect sizes for each categorical predictor against the target. An internal consistency test regressed the composite `existing_controls_score` on its two stated constituent components, testing a relationship asserted by the dataset's own structure.

Predictive modelling applied four algorithms — linear regression, decision tree, random forest and extreme gradient boosting — to four target variables under five-fold cross-validation, yielding sixteen configurations. Algorithms were selected to span a range of functional flexibility. In every configuration the target was excluded from the predictor set to prevent leakage; an initial run omitting this precaution returned an implausible R^2 of 1.000 and was discarded. Performance was benchmarked against predicting the training-fold mean.

Four confirmatory diagnostics followed: permutation-based feature importance; a label-permutation test comparing performance on genuine targets against ten replications with shuffled targets; a learning curve fitted at training sizes from 400 to 8,000 observations; and a classification reformulation in which the target was discretised into three balanced severity bands.

3.5 Tools and Reproducibility

All analysis was conducted in R version 4.5.1, using the randomForest, rpart, xgboost, caret, stringr, dplyr, tidyr, ggplot2 and corplot packages. R was selected for the statistical rigour of its regression and hypothesis-testing facilities, which were central to the diagnostic phase. Python was used at the proposal stage for preliminary exploratory analysis using pandas, matplotlib and seaborn.

A single random seed (40478904) governs all sampling, partitioning and model fitting, rendering every reported result exactly reproducible. Statistical tests used the complete 50,000 records; predictive modelling used a fixed random subsample of 10,000 with capped tree depth, a constraint imposed by available memory. The analysis script and session information are provided in the appendices.

3.6 Ethical Considerations

The study uses publicly available secondary data only and involves no primary data collection. The dataset contains no personally identifiable information and no records relating to identifiable organisations. Data was stored on a password-protected device throughout and will be deleted upon confirmation of the final module mark. The completed ethics approval form was submitted with the project proposal.

One ethical consideration arises from the findings themselves: the study concludes that a publicly available dataset is unfit for its apparent purpose. The analysis is therefore confined to demonstrable properties of the data, with no inference drawn regarding the intentions of whoever produced or published it.

3.7 Methodological Limitations

Four limitations of the design should be noted. Predictive modelling used a 10,000-record subsample rather than the full dataset; all statistical tests, however, used all 50,000 records and returned consistent results. Hyperparameter optimisation was limited to conservative defaults with capped tree depth rather than systematic grid search. The Kolmogorov–Smirnov test issued ties warnings for several variables, which motivated the parallel chi-square procedure. Finally, the inability to verify provenance means that statements regarding generation methodology are inferences from statistical evidence rather than verified descriptions of process.

4. Findings

Statistical tests in Sections 4.1 to 4.4 were conducted on all 50,000 records; predictive modelling in Section 4.5 used a fixed random subsample of 10,000. Interpretation is reserved for Chapter 5.

4.1 Data Quality Assessment

The raw file contained 50,000 records across 22 columns with no missing values and no duplicate records. On conventional measures, the dataset presented as complete and well formed.

Structural inspection revealed that seven columns contained nested JSON objects rather than scalar values; parsing these recovered eleven additional numeric variables, expanding the feature space from 22 to 33 columns. An analyst working from the surface structure alone would have modelled approximately one third fewer variables than the dataset contains.

Three defects were identified. The variable `historical_incident_count` is constant at unity across all records, exhibiting zero variance. Two records report probability values exceeding 1.0, with a maximum of 8.552 — invalid by definition. Twelve columns exhibit unique values in over 90 per cent of records, a cardinality pattern consistent with continuous random generation. Table 4.1 summarises these results.

Table 4.1. Data quality assessment summary
(Source: Author's own analysis, 2026)

Quality check	Result
Total records	50,000
Columns, raw file	22
Columns after JSON extraction	33
Duplicate records	0
Numeric columns with missing values	0
Constant (zero-variance) columns	<code>historical_incident_count</code>
Columns with >90% unique values	12
Invalid probability values (>1.0)	2

Quality check	Result
Maximum observed probability	8.552

4.2 Distributional Analysis

Descriptive statistics revealed marked uniformity across conceptually distinct variables. The target, `threat_intelligence_score`, has a mean of 0.750 and standard deviation of 0.144, bounded between 0.500 and 1.000. Four risk variables, the composite controls score and its two components all exhibit means between 0.749 and 0.751 with standard deviations between 0.144 and 0.145 — nine variables measuring different constructs sharing summary statistics to three decimal places.

Figure 4.1 presents empirical distributions of the six principal bounded variables against the count expected under uniformity. Each histogram is visually flat, oscillating narrowly around the expected 2,000 observations per bin; reduced terminal bins are an artefact of bin boundaries falling on variable limits.

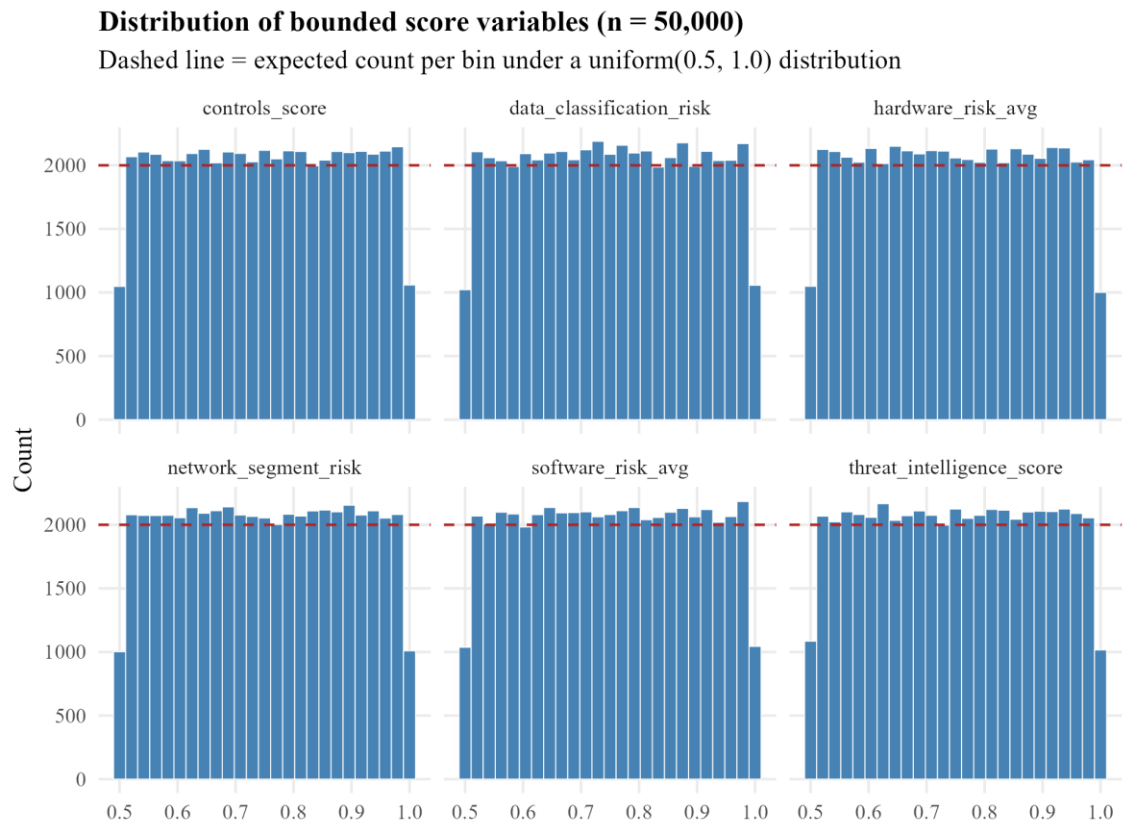


Figure 4.1. Distribution of bounded score variables (n = 50,000)

(Source: Author's own analysis, 2026)

Uniformity was tested using Kolmogorov–Smirnov and chi-square goodness-of-fit procedures (Table 4.2). For eleven of twelve variables neither test rejects uniformity at the five per cent level, and all D statistics fall below 0.006. The exception, *avg_days_since_last_similar*, marginally fails the chi-square test ($p = 0.048$) whilst passing Kolmogorov–Smirnov ($p = 0.252$); across twelve tests, one marginal rejection is consistent with chance.

Table 4.2. Uniformity test results
(Source: Author's own analysis, 2026)

Variable	Range	KS D	KS p	χ^2 p	Uniform?
threat_intelligence_score	0.5–1.0	0.0032	0.702	0.658	Yes
hardware_risk_avg	0.5–1.0	0.0031	0.737	0.865	Yes
software_risk_avg	0.5–1.0	0.0041	0.377	0.585	Yes
network_segment_risk	0.5–1.0	0.0028	0.816	0.531	Yes
data_classification_risk	0.5–1.0	0.0044	0.294	0.429	Yes
controls_score	0.5–1.0	0.0041	0.357	0.968	Yes
preventive	0.5–1.0	0.0033	0.643	0.295	Yes
detective	0.5–1.0	0.0039	0.432	0.209	Yes
prob_occurrence	0.0–1.0	0.0033	0.647	0.509	Yes
avg_days_since_last_similar	30–364	0.0045	0.252	0.048	No
revenue_per_hour	1000–10000	0.0042	0.339	0.076	Yes
data_value_per_gb	500–2000	0.0057	0.077	0.358	Yes

Figure 4.2 illustrates this for the target variable. The empirical cumulative distribution function tracks the theoretical uniform function so closely that the dashed reference line is obscured across the entire range.

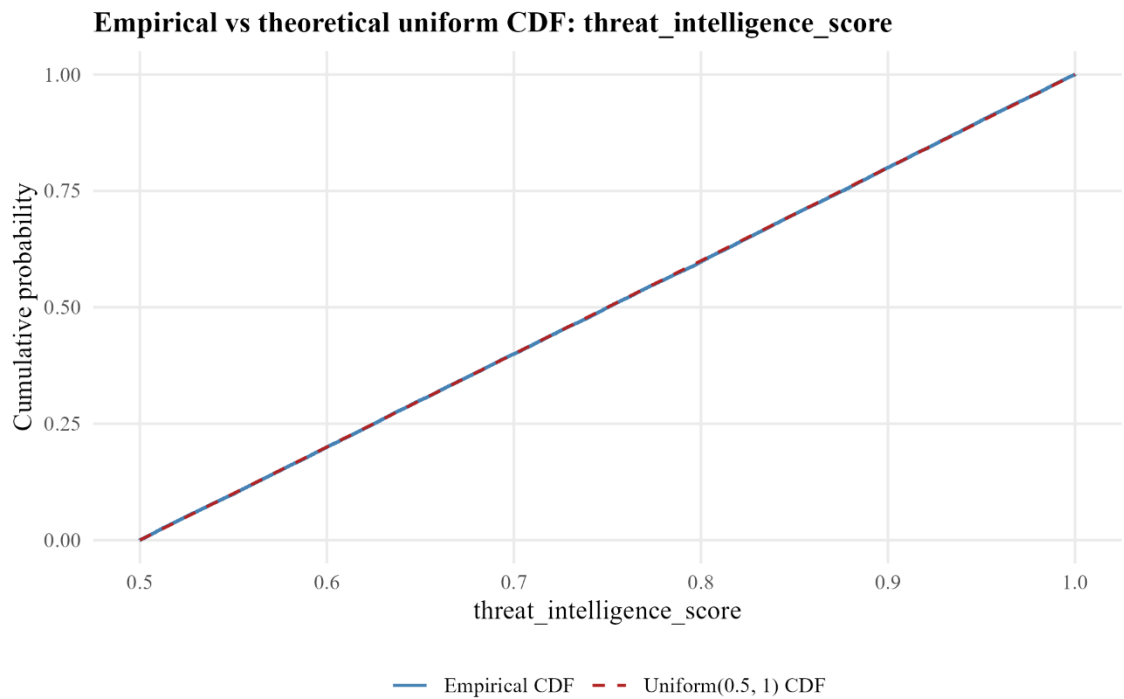


Figure 4.2. Empirical versus theoretical uniform CDF, threat_intelligence_score
(Source: Author's own analysis, 2026)

4.3 Correlation and Association Analysis

Pearson correlations were computed across all fifteen numeric variables with non-zero variance, yielding 120 unique pairwise coefficients (Table 4.3). Not one attains an absolute value of 0.05, conventionally the floor of even a negligible association. The largest observed relationship is $r = 0.0112$; the strongest relationship between the target and any predictor is $r = 0.0092$.

Table 4.3. Summary of pairwise correlations among numeric variables
(Source: Author's own analysis, 2026)

Measure	Value
Unique variable pairs	120
Largest absolute correlation	0.0112
Mean absolute correlation	0.0037
Pairs with $ r > 0.05$	0
Largest absolute correlation with target	0.0092

Pearson correlation matrix (all numeric variables, n = 50,000)

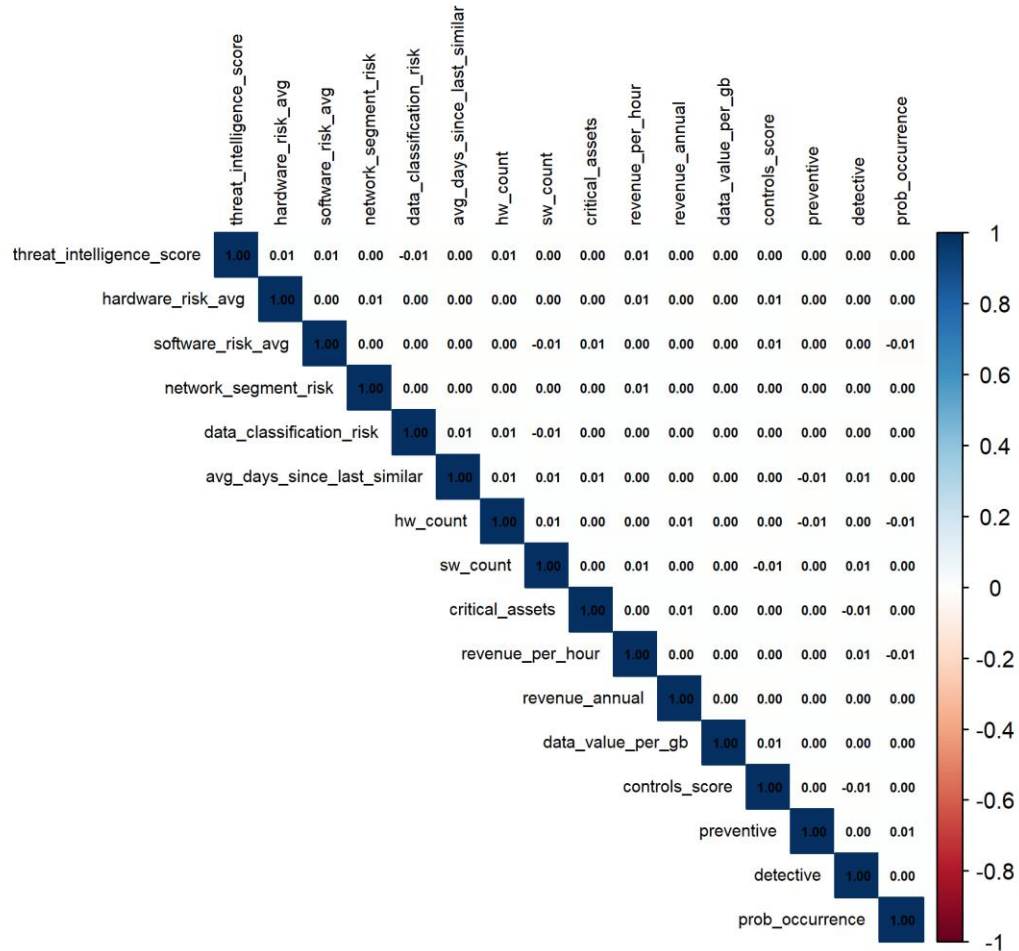


Figure 4.3. Pearson correlation matrix, all numeric variables (n = 50,000)

(Source: Author's own analysis, 2026)

Figure 4.3 renders this visually. With the colour scale fixed at the full theoretical range, every off-diagonal cell appears unshaded — a clean unit diagonal against a blank field.

One-way analysis of variance extended the assessment to categorical predictors (Table 4.4). No categorical variable approaches significance. The largest effect size, for industry_sector, is $\eta^2 = 0.000118$ — approximately one hundredth of one per cent of variance explained. Threat type, the variable most obviously expected to bear on severity, returns $F = 0.105$ and $p = 0.996$.

Table 4.4. One-way ANOVA of target variable by categorical predictor

(Source: Author's own analysis, 2026)

Categorical variable	Levels	F	p	η^2
scenario_type	2	0.001	0.969	0.000000
threat_type	7	0.105	0.996	0.000013

Categorical variable	Levels	F	p	η^2
attack_vector	4	0.938	0.422	0.000056
attack_complexity	4	0.209	0.890	0.000013
initial_access_method	3	0.178	0.837	0.000007
industry_sector	5	1.472	0.208	0.000118

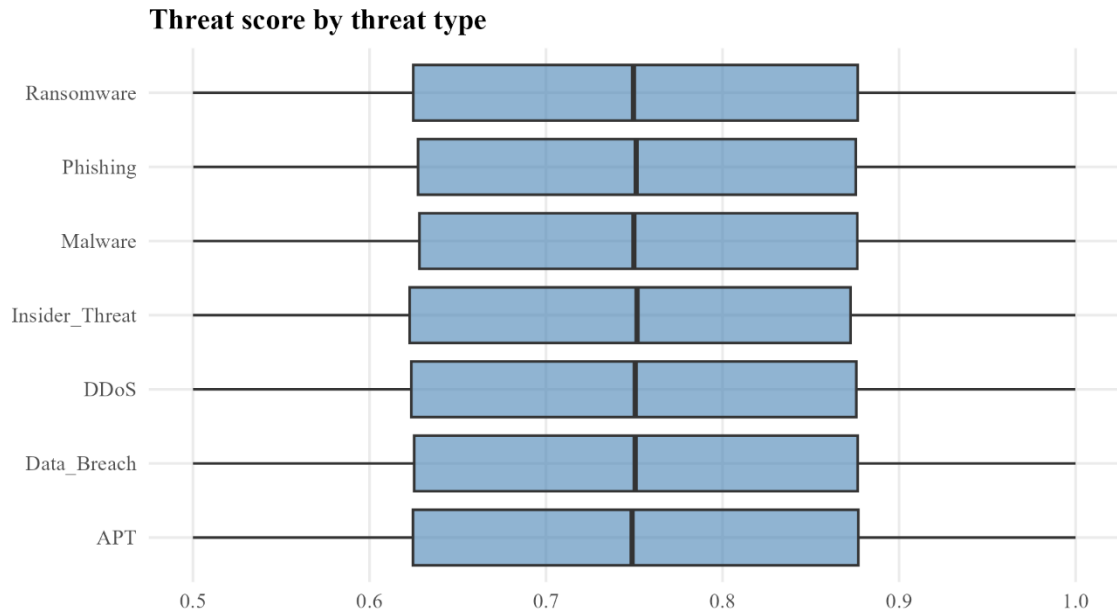


Figure 4.4. Threat intelligence score by threat type
(Source: Author's own analysis, 2026)

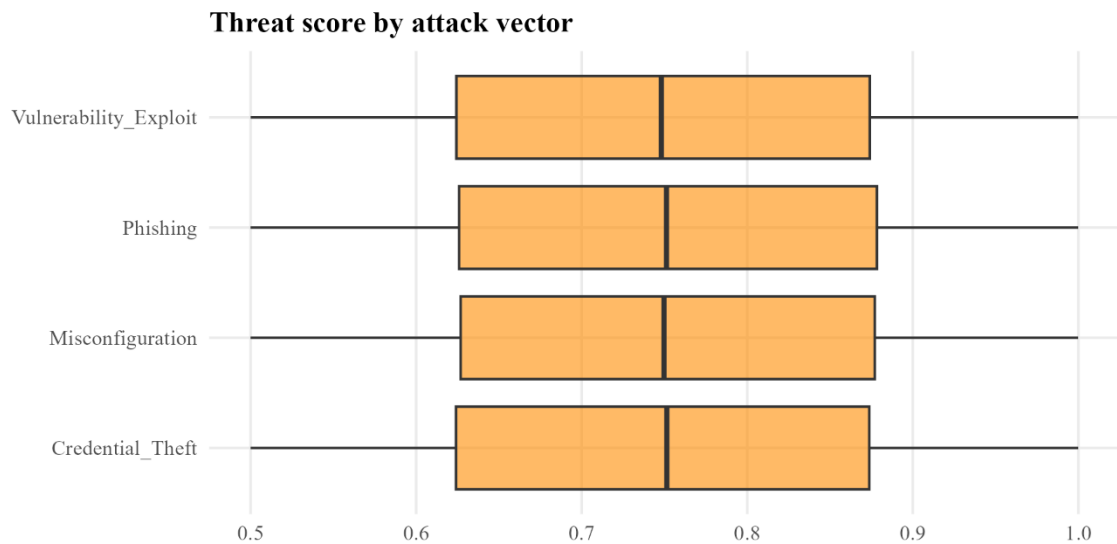


Figure 4.5. Threat intelligence score by attack vector
(Source: Author's own analysis, 2026)

Figures 4.4 and 4.5 display these distributions. Across all seven threat categories and four attack vectors the boxplots are near-identical: medians at approximately 0.75,

interquartile ranges spanning roughly 0.63 to 0.88, whiskers reaching the variable bounds.

Ransomware scenarios receive the same severity distribution as every other category.

4.4 Internal Consistency Testing

Within the `existing_controls` structure, `existing_controls_score` is positioned as a composite with preventive and detective nested beneath as its constituent categories. If internally coherent, the composite would be a function of its components. This was tested by ordinary least squares regression across all 50,000 records (Table 4.5).

Table 4.5. OLS regression of composite controls score on its stated components

(Source: Author's own analysis, 2026)

Term	Estimate	Std. error	t	p
Intercept	0.75825	0.00480	157.92	<0.001
preventive	−0.00304	0.00448	−0.68	0.498
detective	−0.00703	0.00450	−1.56	0.118
R ²	0.000058	—	—	—
Adjusted R ²	0.000018	—	—	—
F-test p-value	0.234	—	—	—

The components jointly explain 0.006 per cent of variance in the composite. Neither coefficient is significant, both carry negative signs, and the overall model is not significant (F-test $p = 0.234$). The intercept of 0.758 is effectively the sample mean: the model predicts the average regardless of component values.

Composite controls score vs its stated components

OLS R-squared = 0.00006 | no coefficient significant at 5%

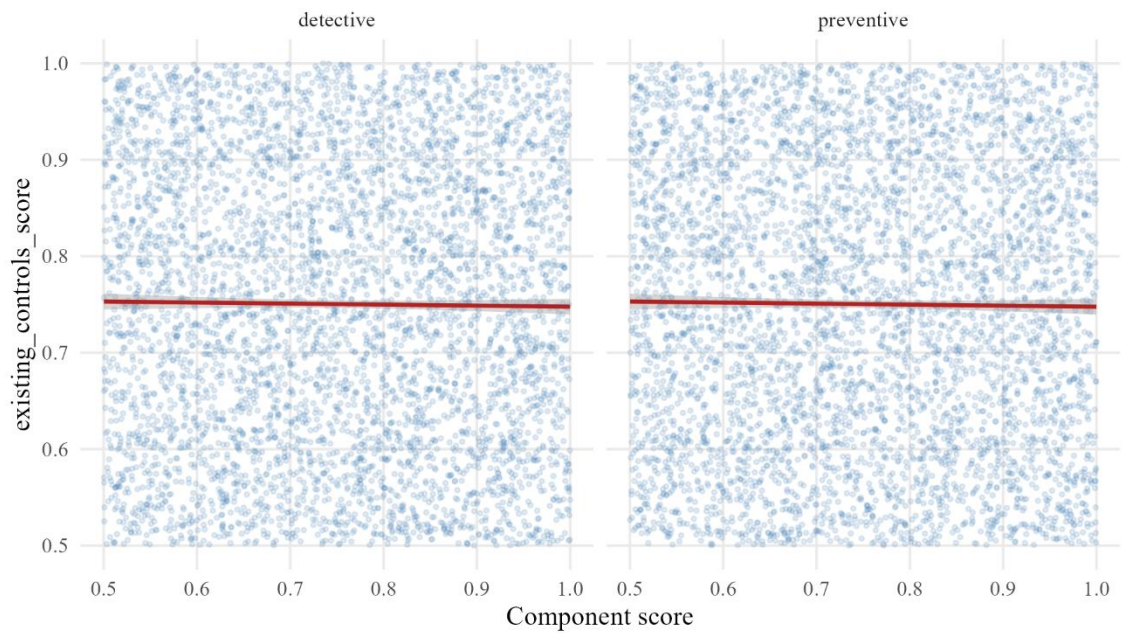


Figure 4.6. Composite controls score versus its stated components

(Source: Author's own analysis, 2026)

Figure 4.6 presents this graphically. Both panels show unstructured scatter filling the bivariate space, with visually horizontal fitted lines. No pattern links a stated summary measure to the values it purports to summarise.

4.5 Predictive Modelling

Four algorithms were applied to four candidate targets under five-fold cross-validation, giving sixteen configurations. Targets were excluded from their own predictor sets. Performance was benchmarked against predicting the training-fold mean, which returns $R^2 = 0$ by construction (Table 4.6).

Table 4.6. Five-fold cross-validated R^2 by target variable and algorithm

(Source: Author's own analysis, 2026)

Target variable	Linear reg.	Decision tree	Random forest	XGBoost
threat_intelligence_score	-0.0039	-0.0003	-0.0064	-0.0184
prob_occurrence	-0.0027	-0.0001	-0.0040	-0.0179
controls_score	-0.0036	-0.0002	-0.0081	-0.0218
critical_assets	-0.0049	-0.0014	-0.0074	-0.0184

All sixteen configurations return negative R^2 ; no model on any target outperforms the naive baseline. Two patterns warrant note. The decision tree produces values closest to zero (-0.0001 to -0.0014), having identified no split worth making and therefore effectively reproducing the mean. Conversely, extreme gradient boosting performs worst on every target. The ordering of algorithms by performance is the exact inverse of their flexibility.

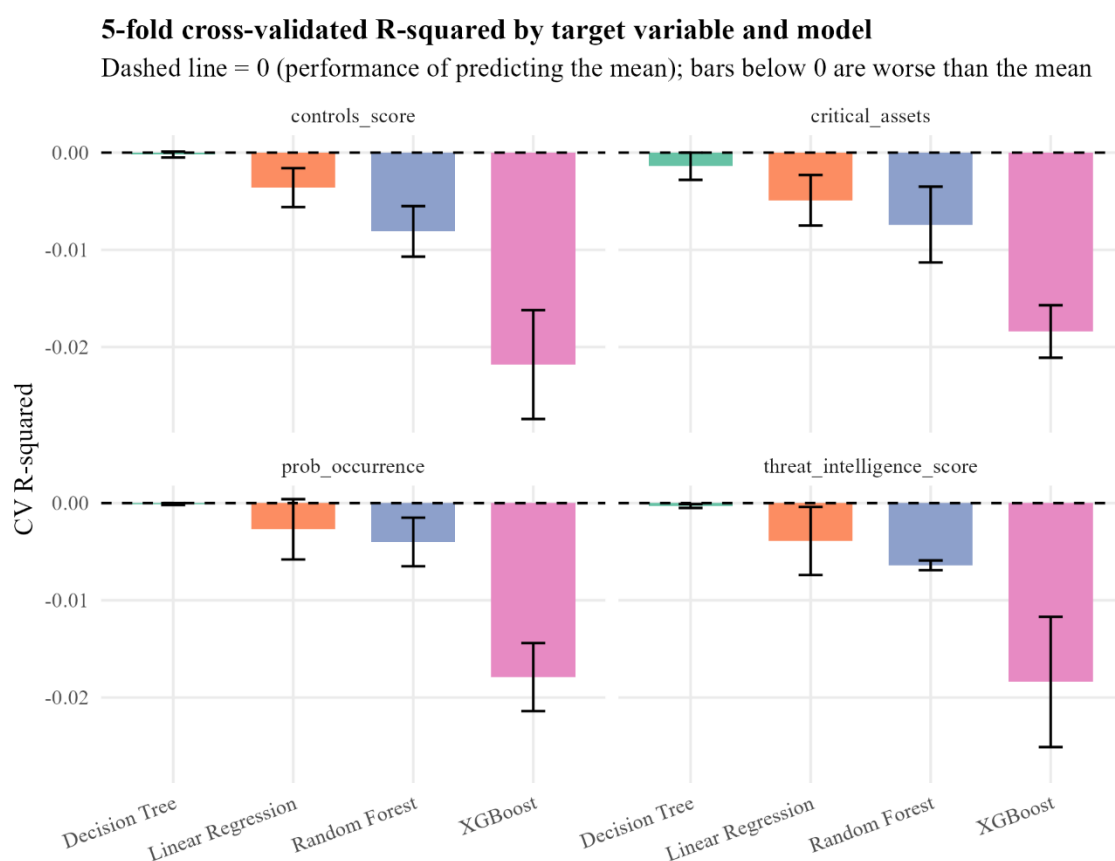


Figure 4.7. Five-fold cross-validated R^2 by target variable and algorithm
(Source: Author's own analysis, 2026)

Permutation-based feature importance was computed from a random forest fitted to the primary target using all seventeen predictors (Figure 4.8). The highest-ranked feature, `hardware_risk_avg`, records a 2.99 per cent increase in mean squared error when permuted. More diagnostically, eight of seventeen features record negative importance, the lowest being `software_risk_avg` at -2.21 per cent — indicating the model performs better when those variables are randomly shuffled than when their true values are supplied.

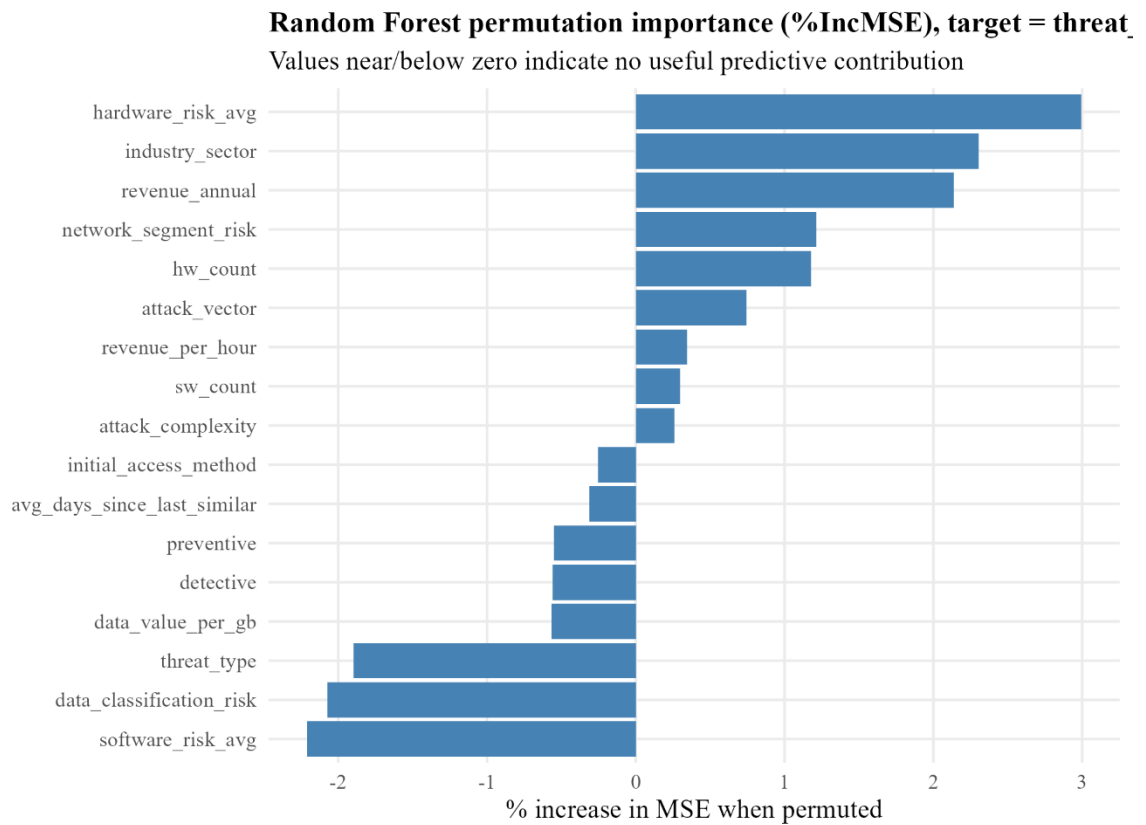


Figure 4.8. Random forest permutation importance, target = threat_intelligence_score
(Source: Author's own analysis, 2026)

A label-permutation test compared performance on genuine targets against ten replications with shuffled targets. The model fitted to the true target achieved $R^2 = -0.0038$; the mean across shuffled replications was -0.0055 (SD 0.0021), giving a permutation p-value of 0.20 (Figure 4.9). Performance on genuine targets is statistically indistinguishable from performance on randomly reassigned labels.

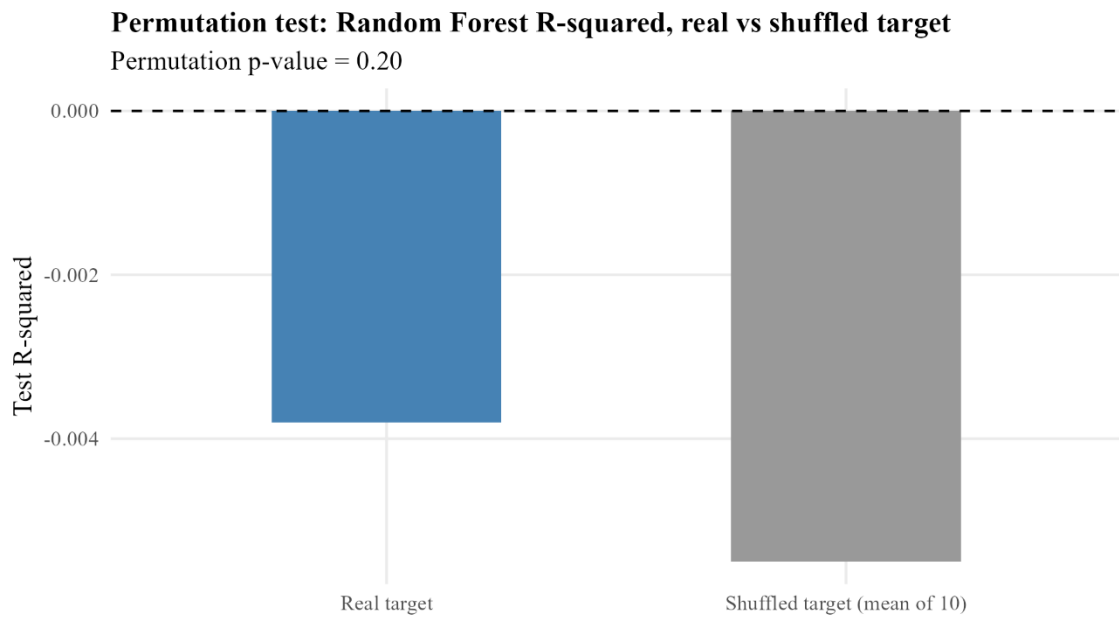


Figure 4.9. Permutation test: random forest R^2 on real versus shuffled targets
(Source: Author's own analysis, 2026)

A learning curve tested whether data volume constrained performance (Figure 4.10). Across training sets from 400 to 8,000 observations, test R^2 fluctuates between -0.045 and -0.004 without crossing zero, the variation consistent with sampling noise rather than systematic improvement.

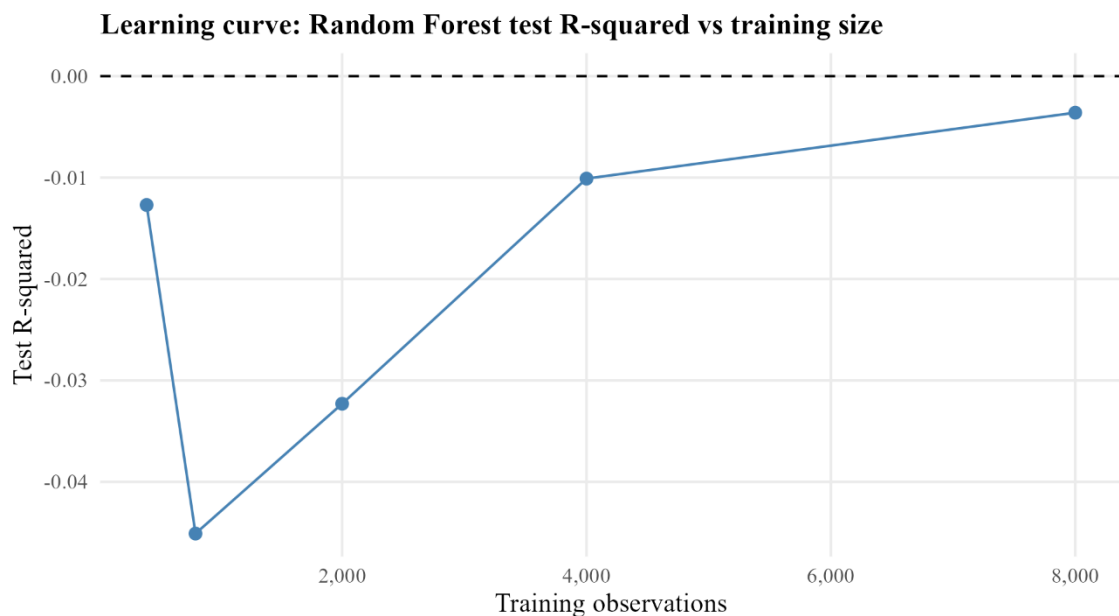


Figure 4.10. Learning curve: random forest test R^2 versus training set size
(Source: Author's own analysis, 2026)

Finally, the target was discretised into three balanced severity bands at its tertiles and a random forest classifier fitted (Table 4.7). The classifier achieved 32.10 per cent

accuracy, below both the 33.33 per cent expected by chance and the 34.45 per cent obtained by always predicting the majority class. Reformulating the task as classification does not recover performance.

Table 4.7. Classification re-framing results (three severity bands)
(Source: Author's own analysis, 2026)

Metric	Value
Random forest three-class accuracy	32.10%
Majority-class baseline	34.45%
Chance accuracy (three balanced classes)	33.33%

4.6 Summary of Findings

The findings converge. Twelve of thirteen numeric variables are statistically indistinguishable from uniform distributions; no pairwise correlation exceeds 0.0112; no categorical variable explains more than 0.012 per cent of variance; and a composite score is explained by its own components to a degree of 0.006 per cent. All sixteen model configurations return negative cross-validated R^2 , eight of seventeen features carry negative importance, performance on genuine targets is indistinguishable from performance on shuffled labels, additional data does not help, and classification reformulation performs below chance. No predictive signal was detected between any variable in this dataset and any candidate target.

5. Discussion of Findings

These findings present an apparent paradox. The same techniques the literature reports achieving accuracies above 95 per cent on cybersecurity data produced no detectable predictive performance here, and a model fitted to randomly shuffled labels performed no worse than one fitted to genuine data. This chapter considers why, and advances the study's principal contribution.

5.1 Why the Models Failed

5.1.1 Excluding methodological explanations

Arp et al. (2022) identify ten recurrent pitfalls in machine learning security research, several of which could in principle produce a misleading null result. The design addressed each.

Model capacity does not appear to be the constraint. Four algorithms spanning a wide range of flexibility were applied; had the relationship been non-linear, the ensemble methods would have detected what the linear model could not. The opposite ordering was observed, with performance declining as flexibility increased — a flexible learner presented with genuine structure exploits it, whereas presented with noise it fits training-fold idiosyncrasies that do not transfer.

Target specification, data volume and feature availability appear similarly not to be constraints. Four dependent variables spanning different scales all produced negative R^2 under all four algorithms; discretisation into severity bands produced accuracy below chance; the learning curve shows no improvement across a twentyfold increase in training size; and expanding the feature set by fifty per cent did not help.

The permutation test provides the strongest evidence against these explanations. Were any operative, a model fitted to true targets would be expected to retain some advantage over one fitted to randomly reassigned labels. None was observed ($p = 0.20$). The

reasonable interpretation is that the models extracted no usable information from the dependent variable, rather than that they extracted it inefficiently.

5.1.2 The character of the data

Three findings, taken together, are consistent with a dataset in which fields were generated independently of one another.

The first is distributional. Twelve of thirteen numeric variables proved statistically indistinguishable from uniform distributions under two independent procedures. Uniformity of this precision is unusual in measured phenomena, since organisational risk scores arise from processes that concentrate observations in some regions and sparsify others. A flat distribution across the entire support is consistent with pseudo-random generation, though it does not by itself establish it.

The second is relational. No pairwise correlation attains 0.0112, and no categorical predictor explains more than 0.012 per cent of variance. Operational data ordinarily exhibits collinearity, because the processes generating it are causally connected; its near-total absence suggests these variables do not share a common generating process.

The third finding is the most informative, because it does not rest on inference from statistical patterns alone. The dataset's schema presents `existing_controls_score` as a composite, with preventive and detective nested beneath it as constituent categories, implying that the summary should be some function of what it summarises. Regression of the composite on its components explains 0.006 per cent of variance and returns two insignificant, negatively signed coefficients. This is not a weak relationship but the apparent absence of one the data structure itself asserts. The most plausible explanation is that composite and components were generated separately. Where the first two findings indicate the data behaves as though randomly generated, this provides more direct evidence that its declared semantics do not match its numerical content.

5.2 Situating the Findings in the Literature

5.2.1 The performance gap

The significance of these results derives from their contrast with published work (Table 5.1).

Table 5.1. Reported performance on authenticated datasets versus the present study
(Source: Author's own analysis, 2026)

Study	Dataset	Principal algorithms	Reported performance
Dasari & Kaluri (2024)	CICIDS2017	LightGBM, XGBoost, RF	99.77% accuracy
Mehmood et al. (2023)	CERT insider threat	LightGBM, XGBoost, RF	97.0% accuracy
Karunamurthy et al. (2025)	MQTT benchmark	Federated deep learning	95.59% accuracy
Yeboah-Ofori et al. (2021)	Microsoft Malware	LR, SVM, RF, DT	F-score 0.85
Present study	Undocumented synthetic	LR, DT, RF, XGBoost	$R^2 = -0.0039$ to -0.0218

The methodological distance is small. Dasari and Kaluri (2024) applied random forest and XGBoost with hyperparameter optimisation to a benchmark intrusion dataset; Mehmood et al. (2023) compared the same ensemble family under identical conditions. Both exceeded 97 per cent. The algorithms here are the same and the validation more conservative. The variable that differs is the data.

The comparison requires care, since those studies address classification whereas this addresses regression, and accuracy and R^2 are not commensurable. The point is directional: where published work demonstrates that ensemble methods extract substantial signal from cybersecurity data, none was detected here under comparable conditions. Zhang et al. (2022) attribute strong performance partly to the availability of high-quality public datasets; this finding is the corollary.

5.2.2 Dataset critique as established practice

A study reporting exclusively negative results invites the objection that it has simply failed; the literature indicates otherwise. Tavallaei et al. (2009) identified redundancy in

KDD Cup 99 that inflated reported performance, prompting construction of NSL-KDD, and later work extended that critique to argue a flawed benchmark shaped two decades of intrusion detection research (Siddique et al., 2019).

This study extends that tradition in two respects. The flaws in KDD Cup 99 were subtle — redundancy and imbalance that inflated performance whilst leaving genuine structure intact, so models trained on it did learn something; here no exploitable structure was detected at all, so the critique concerns fitness for purpose rather than degree of inflation. Second, that tradition diagnoses datasets retrospectively, whereas the protocol in Section 5.4 makes the same diagnosis prospectively at a cost low enough to run routinely.

5.2.3 Data quality, provenance and the cost of neglect

The wider data quality literature explains how such a dataset circulates undetected. Sambasivan et al. (2021) describe data cascades: compounding downstream failures originating in unaddressed upstream problems, arising from a culture that privileges model development over data work. This project is a direct instance — preliminary analysis identified near-zero correlations but recorded them as a modelling risk rather than a data defect, precisely the misclassification their account predicts.

Most directly applicable is Gebru et al. (2021), whose datasheet proposal requires documentation of motivation, composition, collection methodology and intended uses. This dataset satisfies none, and had such documentation accompanied the file its character would likely have been apparent before analysis began. Standards exist and are widely cited (Moustafa and Slay, 2015; Sharafaldin et al., 2018), yet this file carried the same apparent legitimacy as any other publicly hosted dataset.

5.3 The Provenance Problem

Attempts to relocate the original repository listing were unsuccessful; the uploading party, publication date, licence terms and any accompanying description could not be verified retrospectively.

This limitation is substantive rather than procedural. A dataset that cannot be traced to its source cannot be independently validated, nor can findings derived from it be reproduced with confidence that the same data was used. Where no documentation exists, the fitness-for-purpose assessment Gebru et al. (2021) describe becomes impossible in principle.

This bears on the accessibility of public repositories, where the ease of locating analysis-ready files benefits research but, absent any provenance requirement, also permits datasets of unverifiable quality to carry an unearned appearance of legitimacy.

5.4 Contribution: A Synthetic Dataset Validation Protocol

The diagnostic sequence developed here, having proved capable of characterising an undocumented dataset, is offered as the study's principal methodological contribution. It comprises five stages ordered by increasing analytical cost, for execution before substantive modelling begins.

Table 5.2. Proposed synthetic dataset validation protocol
(Source: Author's own analysis, 2026)

Stage	Diagnostic test	Warning indicator
1. Provenance audit	Locate source, author, licence and generation methodology; assess against Gebru et al. (2021) datasheet criteria	Source untraceable or methodology undocumented
2. Distribution screening	Kolmogorov–Smirnov and chi-square goodness-of-fit against uniform distribution for bounded numeric variables	Uniformity not rejected across multiple variables
3. Correlation screening	Full pairwise correlation matrix; ANOVA with eta-squared for categorical predictors	No pairwise $ r $ exceeding 0.05; all η^2 below 0.001
4. Internal consistency	Regress each composite variable on its stated constituent components	Composite not explained by its own components
5. Signal verification	Permutation test comparing performance on true versus shuffled targets	True-target performance not exceeding shuffled baseline

The ordering is deliberate. Stage 1 requires no computation and, applied at the outset, would have raised concern before the dataset was downloaded. Stages 2 and 3 execute in seconds. Stage 4 is the most informative, because it tests a proposition the data itself advances rather than one the analyst imposes. Stage 5 requires model fitting and is placed last.

The protocol's value lies in its economy. Executed in full it requires under an hour of analyst time, against the alternative this project demonstrates: several weeks committed to a dataset incapable of supporting the intended analysis.

5.5 Implications

For research practice, dataset validation should be a distinct phase rather than an implicit component of exploratory analysis. CRISP-DM allocates a data understanding phase but specifies it in terms of description rather than validation. The exploratory analysis at this project's proposal stage met every conventional expectation of that phase and still did not establish that the data was unusable.

For synthetic data, the appearance of quality is not evidence of it. This dataset contains no missing values, no duplicates and perfectly balanced categorical distributions, presenting as exemplary on conventional measures — the same properties independent random generation produces.

For cybersecurity analytics the implications are more serious, since the domain informs the allocation of finite defensive resources. A model trained on data of this character would produce severity scores that appear authoritative whilst bearing little relationship to the risks they purport to quantify. Arp et al. (2022) note that pitfalls in security machine learning often escape detection because reported metrics appear satisfactory; here the failure was total and therefore visible, whereas weak structure would have yielded an apparently successful but equally disconnected study.

5.6 Limitations

Several limitations qualify these conclusions. First, the study establishes that the dataset behaves as though independently generated but cannot establish how it was produced; statements regarding generation methodology are inferences from statistical evidence, supporting the conclusion that the data is unsuitable for modelling rather than any claim about intent.

Second, the analysis concerns a single dataset, so while the protocol is offered as generally applicable, the study supports no inference about the prevalence of comparable deficiencies across repositories. Third, predictive modelling used a 10,000-record subsample owing to memory constraints, though all statistical tests used the complete 50,000 records and returned consistent results.

Fourth, hyperparameter optimisation was limited to conservative defaults; given that no relationship of any magnitude was detected, extensive tuning would amount to optimising against data in which no signal was found. Fifth, the Kolmogorov–Smirnov procedure assumes continuous data without ties, which motivated the parallel chi-square test that returned consistent conclusions in all but one marginal case.

Finally, this study makes no claim about synthetic data as a methodology. Properly constructed synthetic data preserves the statistical relationships of its source; the failure documented here concerns an unvalidated and undocumented instance, not the approach in principle.

6. Conclusions and Recommendations

6.1 Conclusions Against the Objectives

The first objective, to critically review the literature, was met through the four-theme review in Chapter 2, which established both the performance expectation against which this study's results are measured and the tradition of dataset critique within which its contribution sits.

The second objective, to prepare and explore the dataset and assess its suitability, produced the study's pivotal finding: the dataset satisfied every conventional quality criterion — complete, unduplicated, consistently formatted, balanced across categories — whilst proving wholly unsuitable. Conventional quality assessment is necessary but not sufficient.

The third objective, to develop and compare multiple models, was met through sixteen cross-validated configurations. No model outperformed the naive mean-prediction baseline, and performance varied inversely with model flexibility.

The fourth objective, to identify the variables most strongly associated with threat severity, returned a negative answer that is itself the finding. No numeric variable correlates with the target above $r = 0.0092$, no categorical variable explains more than 0.012 per cent of its variance, and permutation importance is negative for eight of seventeen features. The SHAP analysis specified in the proposal was not conducted, since decomposing the predictions of a model without predictive validity would attribute noise rather than signal and would have produced interpretable-looking but meaningless rankings. This is recorded as a deliberate methodological judgement.

The fifth objective, to evaluate performance and discuss practical implications, was met in Chapter 5. The sixth, emerging during execution, was to establish the causes of failure and derive a generalisable validation procedure, producing the protocol in Section 5.4.

The research question asked to what extent machine learning can predict cybersecurity threat severity, and which risk factors most influence it. On this dataset the answer to the first part is that it cannot, and to the second that no factor detectably does. The value of the study lies in the rigour with which those answers were established, and in what establishing them revealed about the data itself.

6.2 Contribution to Practice and Knowledge

The contribution to knowledge is threefold. Empirically, the study provides five independent lines of evidence that a publicly available cybersecurity dataset does not support predictive modelling. The internal consistency test in particular represents a form of evidence not previously applied in this literature: rather than inferring data quality from statistical patterns alone, it tests a mathematical relationship the dataset's own structure asserts, and demonstrates that the relationship does not hold.

Methodologically, the validation protocol operationalises the arguments of Gebru et al. (2021) and Canbek (2022) into an executable sequence of tests. Where those authors argue that documentation and pre-modelling evaluation are necessary, this study specifies what tests to run, in what order, and what results should trigger concern.

For practice, the study demonstrates that the appearance of data quality can be entirely decoupled from its substance, and that the consequences are borne by organisations allocating defensive resources on the basis of arbitrary outputs.

6.3 Recommendations

6.3.1 For analytics practice

Analysts should incorporate the five-stage protocol as a mandatory phase preceding model development, distinct from exploratory analysis. Particular attention should be given to composite variables, whose relationship to their stated components provides the most direct available test of internal coherence. Where provenance cannot be established,

that fact should be recorded as a limitation at the outset rather than discovered at the conclusion.

6.3.2 For data repositories

Public repositories hosting analytical datasets should require, as a condition of publication, structured documentation to the standard proposed by Gebru et al. (2021), specifying at minimum the generation methodology, the provenance of each variable and, for synthetic data, the relationships the generation process was designed to preserve. Repositories should additionally maintain permanent identifiers so that datasets remain traceable after listings are amended or withdrawn.

6.3.3 For future research

Three directions follow. First, the analytical framework specified here should be applied to an authenticated benchmark such as UNSW-NB15 (Moustafa and Slay, 2015) or CICIDS2017 (Sharafaldin et al., 2018), testing the pipeline under known-good conditions and permitting the severity prediction question to be answered substantively. Second, the protocol should be applied systematically across a sample of publicly hosted datasets to establish the prevalence of the deficiencies documented here. Third, the internal consistency test warrants development into a general diagnostic for hierarchically structured data, in which composite variables and their components are identified automatically from schema.

6.4 Concluding Remarks

This study set out to predict cybersecurity threat severity using machine learning. It concludes instead by demonstrating that the data on which such predictions depend cannot be assumed sound merely because it is complete, well formatted and freely available. The techniques applied here are those that, on properly constructed data, achieve accuracies above 95 per cent. Their failure was not a failure of method.

That distinction is the substance of the project. Considerable effort in cybersecurity analytics is directed towards model selection, architecture and tuning, on the implicit assumption that the data is what it claims to be. This study documents a case in which that assumption did not hold, in which no amount of algorithmic capability could compensate, and in which the deficiency was invisible to every conventional quality check applied at the proposal stage. Had the analysis proceeded as originally intended, it would have produced a tuned model, a feature importance ranking and a set of business recommendations, all of them derived from noise and none of them detectably wrong. The validation protocol offered here is a modest safeguard against that outcome, in a domain where the literature suggests the underlying problem is neither rare nor diminishing.

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Appendices

Appendix 1 Dissertation Checklist Sheet

A digitally signed copy of this form should be included as an appendix to the dissertation.

Name: Sanjai Balaji

Date Submitted: 11-09-2026

Signature (Digital): Sanjai Balaji

I confirm that my dissertation contains the following prescribed elements:

- ✓ My dissertation portfolio meets the style requirements set out in the MSc Business Analytics Portfolio Dissertation Handbook including a word count on the front page of each element.
- ✓ I have reviewed the Turnitin similarity report prior to submission.
- ✓ My dissertation title captures succinctly the focus of my dissertation
- ✓ My title page is formatted as prescribed in the MSc Business Analytics Portfolio

Dissertation Handbook

- ✓ The abstract provides a clear and succinct overview of my study
- ✓ Each element contains a Table of Contents, and List of Figures and Tables (where appropriate)
- ✓ My dissertation contains a statement of acknowledgement (optional)
- ✓ The Introduction section of the research report, at a minimum, covers each of the following issues:
 - Background to/context of the project
 - Research question(s), aim(s) and objectives
 - Why the project is necessary/important
 - A summary of the Methodology
 - Outline of the key findings
 - Overview of chapter structure of the remainder of dissertation
- ✓ The Background section of the research report, at a minimum, covers each of the following issues:
 - Synthesises the key technical literature relating to the topic
 - Synthesises the key theoretical literature relating to the topic
- ✓ The methodology section of the research report:
 - Details the procedures adopted in carrying out the project (e.g. the data source/acquisition, data processing, procedures for maximising rigour and robustness, methods of data analysis etc). Contains ethical considerations and

decisions. This section should not duplicate the technical report, which focuses more on the detailed technical choices and steps.

✓ The findings section of the research report reports the results in detail and provides possible explanations for the various findings

✓ The discussion section of the research report makes appropriate linkages between the findings and the literature reviewed

✓ The conclusions section of the research report includes:

- Conclusions about each research question and/or hypothesis
- General conclusions about the research problem
- Implications for theory, for policy and/or management practice
- Limitations of the research
- Suggestions for practice and future research

✓ The technical report, log book, and reflective discussion have each been included.

✓ The reference list is in alphabetical order and follows the Harvard system

✓ I have signed and dated the Candidate Declaration *programme handbook*.